



Rushik Sai Jakka

M.Sc. Controls, Microelectronics & Microsystems

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Portfolio: personal-portfolio-Jakka

About Me

Master's student at the University of Bremen with hands-on experience in **robot programming (ROS 2, Python, C++)**, hardware integration, and **lab-based testing** of robotic systems. Works **independently, in a structured and solution-oriented manner**, enjoys experimenting with prototypes and developing new use cases, with a consistent approach to **verifying and documenting** application functionality — eager to take on the challenges of collaborative robotics.

Experience

Student Assistant (HiWi) (BIMAQ, University of Bremen)

01/2026 – Present

- **Sensor Test Campaigns:** Support the preparation and execution of optical sensor measurement tasks; perform **structured documentation** and verification of findings.
- **Data Acquisition & Calibration:** Design and validate camera calibration workflows for accurate spatial mapping and **reproducible data logging**.
- **Technical Troubleshooting:** Evaluate the robustness of vision algorithms through controlled lab experiments and systematic debugging of sensor setups.
- **Algorithm Implementation:** Implement subpixel-level edge detection to improve precision in vision-based measurement systems.

Master Project: Pan-Tilt Robotic Head for Gaze Estimation

11/2025 – Present

2-DOF Robotic Sensor Head – University of Bremen, Dept.03: Mathematics and Informatics (Computer Science)

- **Robot Programming:** Programming a 2-DOF pan-tilt robotic sensor head using **ROS-2** nodes for real-time sensor processing, coordinate transformations, and actuation control.
- **Mechanical Design:** Designing and modelling platform components in **Fusion 360** for iterative prototyping and part fabrication.
- **Hardware Assembly:** Assembling and configuring the 2-DOF platform with **Dynamixel XM430-W350-R** servo actuators and integrating all electrical connections.
- **Sensor Integration:** Integrating the **Intel RealSense D455f** depth camera for real-time RGB-D data acquisition within the Ros-2 ecosystem.
- **Control Tuning:** Design and tune control loops for stable and responsive actuation of integrated hardware.
- **Iterative Testing & Documentation:** Perform **continuous improvement** of prototype iterations through systematic testing, performance evaluation, and structured reporting.
- **Ros-2 Architecture:** Designing publisher-subscriber node communication pipeline for real-time data flow between the RealSense camera and actuation hardware.
- **URDF Modelling:** Developed a **URDF** robot description file defining the pan-tilt platform's links, joints, and sensor mounts for use within the Ros-2 ecosystem.
- **Simulation & Visualisation:** Simulated and validated platform behaviour in **Gazebo** and monitored real-time joint states and TF transforms using **RViz** prior to physical deployment.

Bachelor Capstone: Modular Assistive Mobility System

07/2022 – 05/2023

Adaptable Smart Wheelchair – VNRVJIET, Hyderabad

- **Prototype Development:** Built and maintained a modular hardware architecture integrating microcontrollers, joysticks, and voice command modules.
- **Sensor Setup & Logging:** Integrated microcontrollers with joysticks and sensors; implemented **data logging** for movement analysis.
- **Technical Troubleshooting:** Resolved integration issues between voice command modules and electrical drive systems.

Volunteer Engagement: IEEE Student Branch Chair

08/2022 – 08/2023

VNRVJIET, Hyderabad, India | ~10 hrs/week | Volunteer

- **Interdisciplinary Leadership:** Led technical symposiums and international conferences (ICMACC'22); coordinated between engineering departments.
- **Strategic Initiative:** Founded the IEEE Robotics & Automation Society (RAS) chapter, managing recruitment and technical roadmaps.

Education

Masters of Science04/2024 – Present

Controls, Microelectronics & Microsystems, Dept.01: Electrical Engineering – University of Bremen, Germany

- Core Focus: Robotic Programming with ROS-2, Intelligent Marine Systems, Sensor Measurement, Process Automation.
- Laboratory: Hands-on testing of IMU sensors (Gyroscope, Accelerometer) and thermal analysis of Thermophile sensor.

Bachelors of Technology08/2019 – 08/2023

Electronics & Instrumentation Engineering – VNRVJIET, Hyderabad, India

- Focus: Industrial Instrumentation, Control Systems, Micro-processors and Micro-controllers.
- Laboratory: Data Logging with IoT, Embedded systems and Hardware coding, Python Programming.

Academic & Lab Projects

Autonomous Navigation & Obstacle Avoidance04/2025 – 09/2025

TurtleBot Implementation for Obstacle Avoidance

- Robot Programming: Programmed navigation and obstacle avoidance behaviours using ROS-2 on a TurtleBot platform.
- LiDAR Processing: Process real-time LiDAR data for obstacle detection and spatial awareness.
- Visualization: Use RViz and Gazebo for real-time monitoring of sensor data and robot state.

Technical Skills

- Robot Programming: Ros-2 (nodes, topics, transforms), robot control and actuation, autonomous navigation.
- Programming: Python (advanced), C++ , Linux/Shell scripting.
- Sensors & Hardware: LiDAR, IMU, Camera Calibration, Data Logging, Sensor Fusion.
- Testing & Tools: Git, RViz, Gazebo, OpenCV, Structured Technical Reporting.
- Hardware Skills: System assembly, soldering, multimeters, and troubleshooting.
- CAD: Fusion 360 (mechanical design and prototyping).

Methodologies

- Analytical thinking, problem-solving, and systematic debugging.
- Verbal communication skills, Design thinking: Double diamond, Root-Cause analysis.

Languages

- English: C1
- German: B1
- Telugu: C1 (Mother Tongue)
- Hindi: Beginner